RESEARCH PROJECT GRANT APPLICATION

Academic Standard Template (ERC / Horizon Europe Equivalent)

1. Project Title

Investigation of the Subjective Horizon of Information (SHI) mechanisms and Landauer's principle in optimizing the cognitive load of biological systems (Project SHIFT).

2. Abstract

The project aims to empirically verify the existence of the Subjective Horizon of Information (SHI)—a mechanism by which the human cognitive system dynamically and selectively renders the physical parameters of matter strictly within the boundaries of Direct Spatial Interaction (BIP, 1.5–6 m). Grounded in Predictive Processing and the thermodynamics of information, the study will investigate the energetic cost of purging redundant variables from working memory (in relation to the Landauer limit: $\Delta\mathrm{S} + \Delta\mathrm{I} \geq \mathrm{k}\ \ln\ 2$). Furthermore, the role of neuroacoustics (including generative AI stimulants) in clocking and modulating the capacity of the perceptual horizon will be verified. The findings will enable the development of *Interaction-Driven Culling* algorithms, revolutionizing operator ergonomics and establishing the foundation for a new generation of Edge AI systems.

3. Scientific Objectives

The primary objective is to understand how the brain optimizes data processing under conditions of "repetitive spatiotemporal loops" (e.g., critical system operators, logistics) by suppressing the perceptual resolution of the background.

Specific objectives:

- Define the boundary radius (Markov Blanket) beyond which objects undergo cognitive decoherence.
- Measure the impact of repetitive audio algorithms (e.g., looped audio streams/YouTube) on inducing a "low-resolution rendering" mode in the nervous system.
- Investigate the neurobiological and metabolic costs of intentional working memory resets (*Selective Information Erasure*).

4. State of the Art & Significance

Current artificial intelligence models and classical cognitive science struggle with the combinatorial explosion problem—attempting to process the entire available environment at maximum resolution. In logistics and industry, this leads to human informational overload ("information noise"), resulting in critical errors. Project SHIFT proposes a revolutionary approach based on Digital Ontology: treating consciousness as an energy-efficient "rendering engine." Proving that the human brain naturally employs Interaction-Driven Culling will pave the way for transferring these mechanisms to the architecture of artificial neural networks, solving the computational bottlenecks of modern technology.

5. Concept and Methodology (Work Packages)

- **WP1: VR Behavioral Studies (Defining the BIP Bubble).** Utilizing Virtual Reality headsets with integrated eye-tracking. Subjects will perform repetitive logistical tasks in a loop. We will measure reaction times to appearing/disappearing background objects as a function of distance (0-10m). Factorial ANOVA will be applied to isolate confounding variables and confirm the drop in perceptual coherence beyond the 6m horizon.
- **WP2: Thermodynamics of Perception Measurements (fMRI / EEG).** Analysis of the Default Mode Network (DMN) and Prefrontal Cortex (PFC). Investigating the measurable metabolic expenditure when initiating the working memory clearing procedure (*Selective Information Erasure*), isomorphically referenced to the minimum Landauer limit ($W = kT \ln 2$). Verification of the energetic cost required to return the system to homeostasis.
- **WP3: Neuroacoustic Validation.** Exposing subjects to varied algorithmic stimuli (selected generative AI music vs. algorithmic Lo-Fi loops from video platforms). Measuring the impact of these frequencies on the rate of background object decoherence. Verifying the hypothesis that externally clocked audio stabilizes or disrupts the reality rendering speed.
- **WP4: Algorithm Development (Proof of Concept).** Mathematical synthesis of results from WP1-WP3 into a prototype *Interaction-Driven Culling* algorithm for digital systems (achieving TRL 4).

6. Safety and Research Ethics

All interventions are strictly non-invasive. The *Selective Information Erasure* protocol will be implemented as a form of biofeedback and cognitive hygiene training, without the use of pharmacological agents. Biometric measurements (EEG/fMRI) are standard, safe medical procedures. The project includes rigorous test termination procedures if excessive cognitive stress is detected in participants.

7. Expected Impact & Implementation Potential (ROI)

- **For Science:** Providing hard evidence that resource optimization necessitates the cognitive decoherence of objects. Establishing a new mathematical model linking statistical physics (Landauer's Principle) with neurobiology.
- **For Business and Economy (Commercialization Phase):** The project will deliver a rigorously tested logical model ready for integration with Autonomous Vehicle (AV) and Edge AI systems. Focusing processors on rendering hard physics exclusively within the BIP range creates the potential to reduce LiDAR sensor noise by approx. 30-40%, drastically accelerating AI decision-making.

- **For Society:** Developing Occupational Health and Safety (OHS) guidelines for high-digital-entropy environments, translating to a reduction in routine workplace accidents and protecting operators from cognitive burnout.